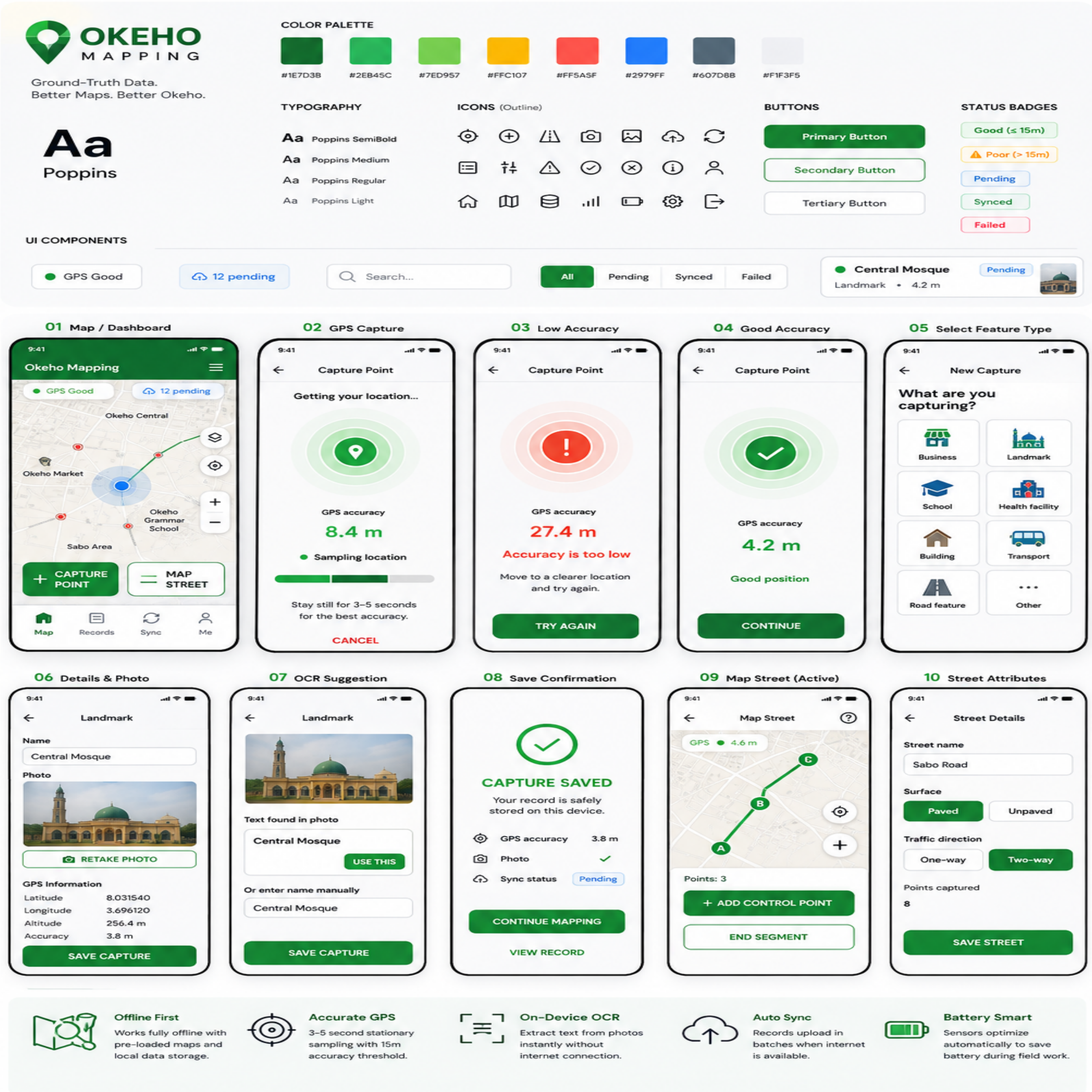
OKEHO MAPPING APP
UI/UX Design & Product Plan

Ground-truth field capture and mapping system
Okeho, Oyo State, Nigeria • August 2026



Purpose: translate the supplied PRD into a coherent, implementable mobile UX—with particular attention to offline-first capture, GPS accuracy, OCR, street mapping, local queues, and synchronization.

1. Product Basis

The supplied PRD defines an offline-first Flutter mobile application for collecting ground-truth mapping data in Okeho. Its purpose is to capture, validate and normalize field data for a PostGIS-backed pipeline and downstream OSM and institutional submission workflows.

The core field workflows are point-feature capture and line-feature/street-segment capture. The PRD also establishes GPS sampling and accuracy thresholds, camera capture, on-device OCR, offline local storage, offline map tiles, cold-start GPS handling, automated synchronization, and GeoJSON-standardized backend records.

Source boundary: authentication, the complete feature-type taxonomy, and some profile/settings details are not fully specified in the supplied PRD. Those areas are therefore marked as proposed UX rather than fixed requirements.

2. UX Strategy

• Capture first, explain second: minimize typing and interaction while workers are in the field.

• Map-first: make the map the primary working surface rather than a secondary screen.

• Offline success is still success: a record saved locally should be clearly confirmed even when there is no network.

• Make sensor state visible: GPS quality, sync state and offline status should be understandable at a glance.

• Use large field controls: primary actions such as Capture Point, Start Segment, Add Control Point and End Segment should be easy to hit outdoors.

• Separate point and street workflows: they share the same system but have different interaction patterns.

3. Information Architecture

Area	Primary purpose
Map	Main field-working surface; view offline map, current location and captured features.
Records	Review captured records and their status: all, pending, synced and failed.
Sync	Show queued uploads, active synchronization, completed syncs and failures.
Profile / Settings	Worker/device and app settings; exact scope to be confirmed.
Capture Point	GPS acquisition → feature type → details/photo → OCR → review → local save.
Map Street	Start segment → control points → end segment → street attributes → save.

Primary navigation

Map → Records → Sync → Profile

Map should be the default destination because the majority of field activity starts from the geographic working surface.

4. Primary User Flows

Point capture

Map → Capture Point → GPS sampling → accuracy check → feature type → details/photo → OCR suggestion → review/save → local queue → later synchronization.

Street capture

Map → Map Street → Start Segment → add control points → End Segment → street name/surface/direction → local save → later synchronization.

5. Screen Inventory

Group	Screens
Setup / proposed	Splash; Login; Field-worker setup; Permissions; Offline map download.
Core	Map dashboard; Capture Point; GPS warming; GPS accuracy warning; Feature Type; Feature Details; Ca
Street mapping	Start Segment; Active Segment; Add Control Point; End Segment; Street Attributes; Street Saved.
Records	All Records; Record Details; Pending Records; Failed Records; Sync.
Settings / proposed	Profile; GPS/device status; Offline Maps; Storage; App Settings.

6. Core Screen Specifications

Map / Field Dashboard

- Map occupies the majority of the screen.
- Show GPS status and sync status near the top.
- Show current position, captured points and mapped streets.
- Primary actions: Capture Point and Map Street.
- Bottom navigation: Map, Records, Sync, Profile.

GPS Capture

- Show that the app is sampling the user's stationary position.
- Display GPS accuracy in metres.
- Use a clear progress/status treatment during the 3–5 second sampling window.
- If accuracy is greater than 15 m, block progression and explain how to improve the position.

Feature Details / OCR

- Name field plus photo capture.
- Show latitude, longitude, elevation and accuracy as system-collected information.
- Show OCR text suggestions beneath the relevant manual field.
- Allow the worker to accept an OCR suggestion with one tap.

Save Confirmation

- Clearly state that the record is saved on the device.
- Show GPS accuracy, photo status and sync status.
- Offer Continue Mapping and View Record.

7. Street Mapping UX

The street workflow should be map-dominant. The worker should be able to see the emerging geometry and the control points while walking the segment.

1. Start Segment
Worker taps Start Segment at the beginning node.

2. Add Control Point
Worker taps at curves, bends, midpoints or other required control locations.

3. End Segment
Worker taps End Segment at the terminal node.

4. Attributes
Enter street name, surface type and traffic directionality.

5. Save
Store the completed segment locally and mark it for synchronization.

8. Offline-First UX

• Offline map tiles remain available for Okeho.

• Capture records, metadata and photo assets are stored in the local queue.

• Cold GPS starts should show a 'Warming up GPS...' state rather than appearing frozen.

• When cellular data or Wi-Fi becomes available, queued records can synchronize automatically.

• The worker should always be able to distinguish saved locally from synced remotely.

Recommended status vocabulary

State	Meaning
Good	GPS accuracy is within the 15 m capture threshold.
Poor	GPS accuracy exceeds the threshold; user should improve position.
Pending	Record is safely stored locally but not yet synchronized.
Syncing	Queued data is currently being uploaded.
Synced	Backend synchronization completed.
Failed	Synchronization failed and requires retry or review.

9. Visual Design System — Initial Direction

The visual direction shown in the UI mockup uses a clean, field-oriented green system with high contrast, large touch targets and restrained decoration. This is a proposed visual direction, not a color specification contained in the PRD.

Role	Proposed value	Use
Primary	#167D3B	Primary actions, navigation emphasis, success states.
Secondary green	#2EB45C	Secondary accents and supporting states.
Light green	#E7ED957	Light backgrounds/accent areas; verify final contrast before implementation.
Warning	#FFC107	Warnings and attention states.
Error	#FF5A5F	GPS failure and sync failure states.
Info	#2879FF	Informational/pending states.
Neutral	#607D8B	Secondary metadata and neutral UI.

Typography proposal: Poppins for headings and interface text, with a clear weight hierarchy.

Component direction: large primary buttons, compact status badges, clear cards, outline icons, high-contrast map controls, and consistent spacing.

10. Component Library

- Primary / secondary / tertiary buttons
- Text inputs and selection controls
- Status badges: Good, Poor, Pending, Syncing, Synced, Failed
- GPS status indicator and accuracy display
- Sync status indicator
- Map control buttons
- Feature-type cards
- Record cards
- Photo capture/review component
- OCR suggestion component
- Confirmation / error / warning banners
- Bottom navigation

11. Required State Matrix

Area	States to design
GPS	Searching; warming up; sampling; good; poor; unavailable.
Network	Online; offline; reconnecting.
Sync	Idle; pending; syncing; completed; failed; retrying.
Point capture	Ready; capturing; review; saved locally; save error.
Camera	Ready; captured; retake; image saved; image error.
OCR	Processing; suggestion found; no text found; suggestion accepted.
Street	Ready; active; control point added; end confirmation; saved locally.
Records	All; pending; synced; failed; record detail.

12. Prototype Plan

1. Design system
Lock typography, colors, spacing, components and status vocabulary.

2. High-fidelity screens
Turn the wireframes into consistent production-oriented mobile screens.

3. Interaction prototype
Connect the primary point-capture and street-mapping flows.

4. State testing
Walk through offline, GPS, OCR, sync and failure states.

5. Developer handoff
Document component behavior, validation, navigation, data dependencies and implementation notes.

13. Technical UX Alignment

The UI should expose only information that helps the field worker make decisions while preserving the required technical metadata in the underlying record.

- Point captures include latitude, longitude, elevation and GPS accuracy.
- The system uses continuous sampling to calculate a representative position.
- Captured images are compressed to reduce local storage and synchronization bandwidth.
- OCR runs on-device and therefore should not be represented as an internet-dependent step.
- Local records form a queue for later synchronization.
- Backend records are normalized into GeoJSON-compatible feature structures.

Example backend fields supplied in the PRD include capture ID, feature type, name, OCR suggestion, GPS accuracy, contributor ID, photo path and timestamp.

14. Scope Boundaries / Decisions Still Needed

- Confirm the complete feature-type taxonomy.
- Confirm authentication and worker-account requirements.
- Confirm whether field workers can edit or delete records after local save.
- Confirm duplicate-detection behavior.
- Confirm exact offline-map download/update workflow.
- Confirm permissions and device-support requirements.
- Confirm sync conflict handling and backend error messages.
- Confirm whether supervisors/admins need a separate workflow or interface.

These items should be resolved before the design is considered final because they can materially change navigation, screens and component behavior.

15. Recommended Next Deliverable

High-fidelity UI + component library + clickable prototype specification.
The immediate focus should be the Map dashboard, GPS capture, point details/OCR, save confirmation, active street mapping and street attributes. After those are approved, the remaining screens can be built from the same component system.